

9. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

- (i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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- (ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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- (b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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- (c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



24

1	H	Hydrogen	1
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7	Li	Lithium	3
9	Be	Beryllium	4
23	Na	Sodium	11
24	Mg	Magnesium	12
39	K	Potassium	19
40	Ca	Calcium	20
86	Rb	Rubidium	37
88	Sr	Strontium	38
133	Cs	Caesium	55
223	Fr	Francium	87
137	Ba	Barium	56
226	Ra	Radium	88
227	Ac	Actinium	89

11	B	Boron	5
12	C	Carbon	6
14	N	Nitrogen	7
16	O	Oxygen	8
19	F	Fluorine	9
20	Ne	Neon	10
27	Al	Aluminium	13
28	Si	Silicon	14
31	P	Phosphorus	15
32	S	Sulfur	16
35.5	Cl	Chlorine	17
40	Ar	Argon	18
70	Ga	Gallium	31
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
128	Te	Tellurium	52
127	I	Iodine	53
131	Xe	Xenon	54
204	Tl	Thallium	81
207	Pb	Lead	82
209	Bi	Bismuth	83
210	Po	Polonium	84
210	At	Astatine	85
222	Rn	Radon	86

59	Co	Cobalt	27
56	Fe	Iron	26
55	Mn	Manganese	25
52	Cr	Chromium	24
51	V	Vanadium	23
48	Ti	Titanium	22
45	Sc	Scandium	21
89	Y	Yttrium	39
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
99	Tc	Technetium	43
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
108	Ag	Silver	47
112	Cd	Cadmium	48
63.5	Cu	Copper	29
65	Zn	Zinc	30

Key

Ar	Symbol	Name	Z
		relative atomic mass	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3430U20-1



Z22-3430U20-1

FRIDAY, 17 JUNE 2022 – AFTERNOON**SCIENCE (Double Award)****Unit 2 – CHEMISTRY 1
FOUNDATION TIER**

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	6	
2.	11	
3.	6	
4.	5	
5.	6	
6.	5	
7.	6	
8.	9	
9.	6	
Total	60	

3430U201
01**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7 is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN223430U20101

9. (a) In 1915 Alfred Wegener suggested that the Earth’s continents were once joined together as one large land mass.



- (i) State **three** pieces of evidence that Alfred Wegener used to support his theory. [3]

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- (ii) Explain why Wegener’s theory was not originally accepted by other scientists, but it is today. [2]

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(b) What type of destructive event is likely to happen at a **conservative** plate boundary? [1]

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Examiner
only

6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
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iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



	Group
1	1
2	2
3	3
4	4
5	5
6	6
7	7
0	0

Key

Symbol	Name	Z	A_r
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Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3430U20-1



FRIDAY, 16 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 2 – CHEMISTRY 1
FOUNDATION TIER

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	9	
3.	9	
4.	8	
5.	6	
6.	5	
7.	8	
8.	7	
Total	60	



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